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Project Name	Smart Lender: Machine Learning-Based Loan Approval Prediction System
Maximum Marks	

Solution Requirements

Introduction

The Solution Requirements define the functional capabilities and expected behavior of the Smart Lender Loan Approval Prediction System. These requirements specify the services provided by the application, user interactions, system operations, data processing, and Machine Learning functionality required to deliver accurate and efficient loan approval predictions.

1. Functional Requirements (FR)

FR ID	Requirement	Description	Priority
FR-OI	User Registration	Applicants shall legister securely using personal information.	High
FR-02	User I.ngin	Registered users shall authenticate using valid credentials.	High
FR-03	Profile Management	Users shall update their personal information.	Medium
FR-04	Loan Application Submission	Applicants shall submit financial, employment, and demographic details.	High
FR-05	Input Validation	system shall validate mandatory' fields before submission.	High
FR-06	Data Preprocessing	Ihe system shall clean and transform applicant data before prediction.	High

	Machine Learning Prediction	System shall predict loan approval or rejection using trained models	High
FR-08	Prediction History	Store previous prediction results.	Medium
FR-09	Adminlstrator Dashboard	Administrators shall monitor applications and pediction reports.	Medium
FR-IO	Report Generation	Generate downloadable prediction reports and analytics.	Medium

Functional Requirements Summary

The functional requiremets define the essential services required for secure user management, application ptocessing, intelligent prediction, leponing, and system administration.

2. Non-Functional Requirements (NFR)

NFR ID	Requirement	Description	Acceptance Criteria
NFR-OI	Performance	mediction shall be generated quickly.	Response time less than 2 seconds.
NFR-02	Accuracy	Machine I_æaming model shall pmvide reliable predicuons.	Prediction accuracy gæater than 90%.
NFR-03	Security	Pmtect user information using secure authentication and encryption.	HITPS, encrypted passwords, role—based access.
NFR-04	Availability	Application shall temain continuously available.	99.9% uptime.
NFR-05	Scalability	Support increasing numlrrs of users.	Minimum registered users.
NFR-06	Usability	Provide a simple and intuitive interface.	Easy navigation and responsive design.
	Reliability	Ensure stable system operation.	Minimal prediction failures.
NFR-08	Maintainability	Support future enhancements and model updates.	Modular architecture.

3. System Assumptions

- Users provide accurate and complete applicant information
- Internet connectivity is available for system access
- Trained Machine Learning model is deployed successfully
- Database server operates reliably
- Only authorized users access administrative features

4. System Constraints

- Prediction quality depends on the quality of training data
 - Internet connection is required for system operation
 - Incomplete or incorrect input may affect prediction accuracy
 - System performance depends on available server resources
 - Model retraining requires updated datasets
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Conclusion

The Solution Requirements define the functional and non-functional expectations of the Smart Lender Loan Approval Prediction System. These requirements ensure that the system remains secure, scalable, reliable, user-friendly, and capable of delivering accurate loan approval predictions using Machine Learning techniques.